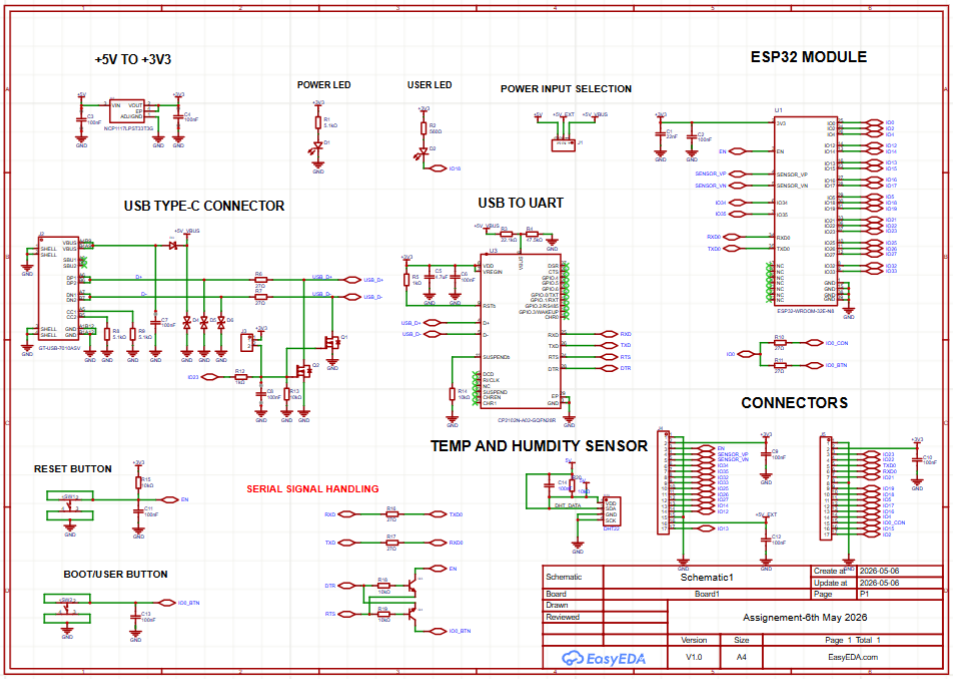
PART B

Number 2



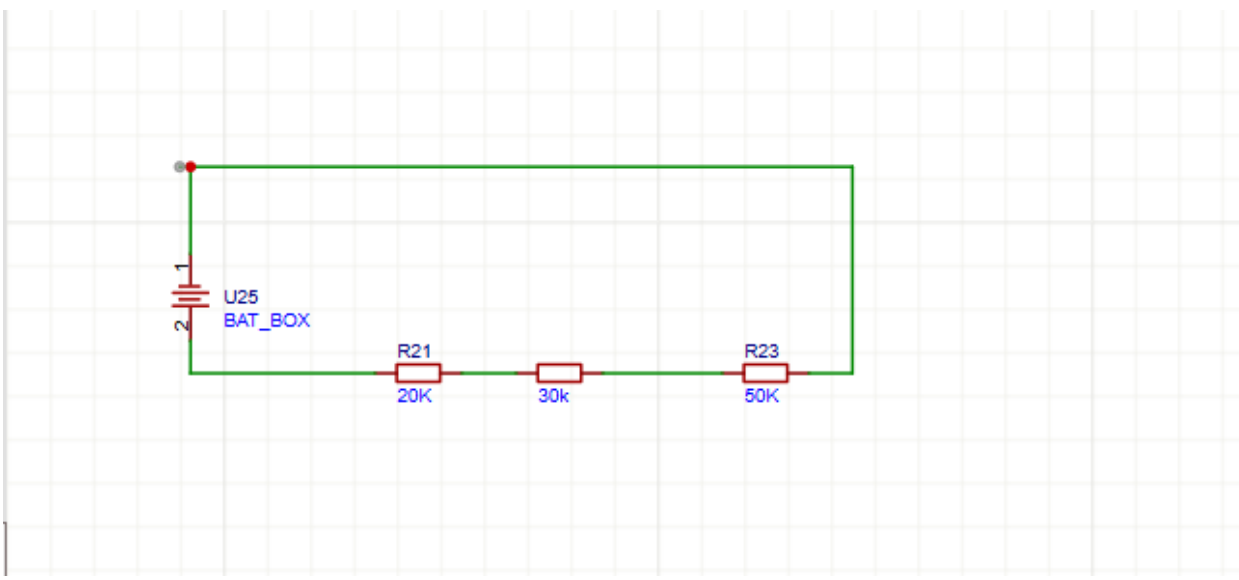
Table

Number	Reference	Component	Value/Model	Quantity
1	U1	Microcontroller Module	ESP32-WROOM-32E-N8	1
2	U2	USB to UART Bridge	CP2102N-A02-GQFN28R	1
3	U3	Voltage Regulator	NCP1117LP3233T3G	1
4	U4	Temperature & Humidity Sensor	DHT22	1
5	J1	USB Type-C Connector	GT4USB-7010ASV	1
6	J3/J4	GPIO Connectors	Pin Headers	2
7	R1	Pull-up Resistor (DHT22 Data)	10kΩ	1
8	R2	Current Limiting Resistor (Power LED)	1kΩ	1

9	R3	Current Limiting Resistor (User LED)	1kΩ	1
10	D1	Power LED	LED	1
11	D2	User LED	LED	1
12	SW1	Reset Button	Tactile Push Button	1
13	SW2	Boot/User Button	Tactile Push Button	1
14	C1-C4	Decoupling Capacitors (+5V TO +3V3 & ESP32)	100nF	4
15	C5	Filter Capacitor (Voltage Regulator output)	10µF	1

Number 1

Circuit A

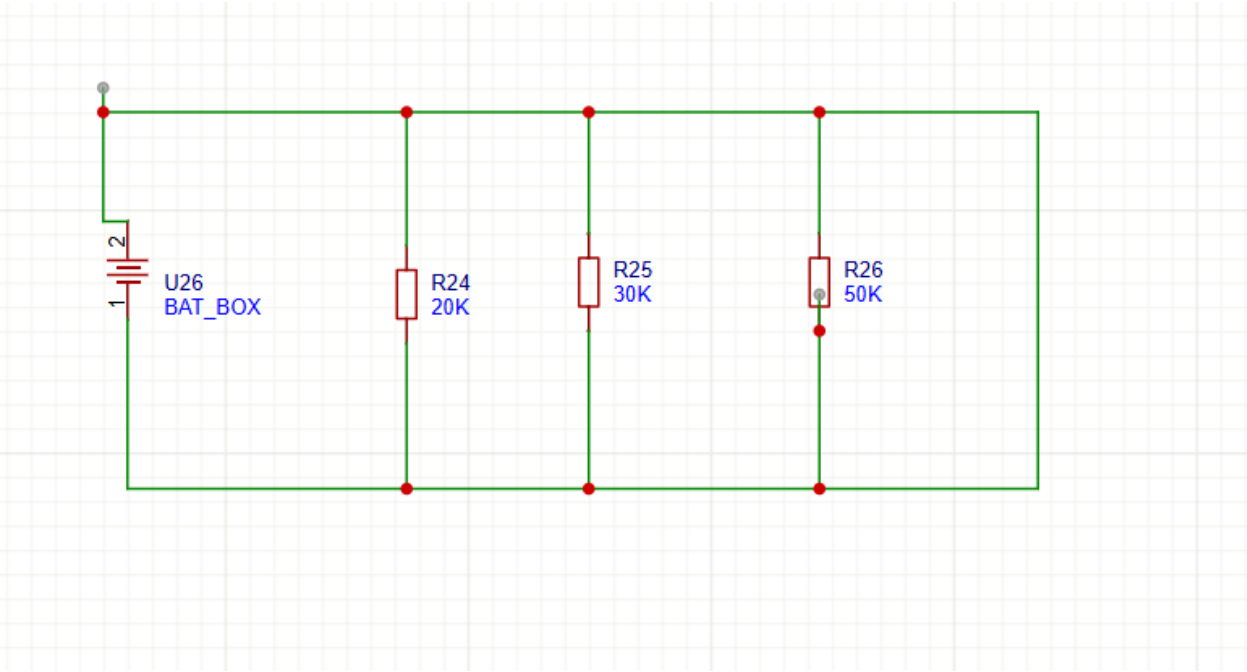


Table

Number	Reference	Component	Value	Quantity
1	U25	DC Power Supply	125V	1
2	R21	Resistor	20Ω	1
3	R22	Resistor	30Ω	1

4	R23	Resistor	50Ω	1

Circuit B



Table

Number	Reference	Component	Value	Quantity
1	U26	DC Power Supply	125V	1
2	R24	Resistor	20Ω	1
3	R25	Resistor	30Ω	1
4	R26	Resistor	50Ω	1